



前沿探索与大作业安排 (2025)

李文根/Wengen Li

Email: lwengen@tongji.edu.cn

先进数据与机器智能系统实验室

Advanced Data and Machine Intelligence Systems (ADMIS) Lab

<https://admis-tongji.github.io>

同济大学 计算机科学与技术学院

2025年04月

- **方向1: 蚂蚁科技OceanBase、华为openGauss/GaussDB (选择下列某个方面)**
 - Storage: 存储
 - Index: 索引
 - Query: 查询
 - Optimization: 优化
 - Transactions: 事务处理、并发、恢复
- **方向2: 多模态数据检索(Multimodal Data Retrieval)**
- **方向3: 向量数据库(Vector Database)**
- **分组**
 - 建议每组1-3人 (如果超过3人应与课程老师或助教单独沟通)
 - Choose one topic and search for related materials
 - Prepare slides (20-30 pages) and give the presentation
 - 内容完整, 逻辑性强; 具有独立的思考; 页面美观; 个人分工明确
 - **第14周提交**

• 数据库应用开发选题要求

- 围绕“**海洋计算**”或“**城市计算**”的合适题目
- **海洋计算**：海洋遥感图像检索系统、海洋表面温度预测平台、碳酸盐补偿深度CCD空间预测系统、船舶轨迹查询系统、台风监测系统、港口集装箱管理平台等
- **城市计算**：城市交通状态查询系统、城市犯罪事件管理平台、城市基层社会治理知识图谱管理等
- 题目自拟，建议具有一定的新颖性，**4月21日**前完成，在Canvas上录入

• 个人单独完成，要求涵盖以下内容：

- 需求分析
- 概念设计（E-R图）
- 逻辑设计，要求数据库各个关系满足3NF
- 物理设计，简要阐明如何构建索引结构
- 系统实现（下学期课设完成，本学期不要求）

• 提交一份10页以上，不超过20页（包括封面、目录和参考文献）的大作业报告

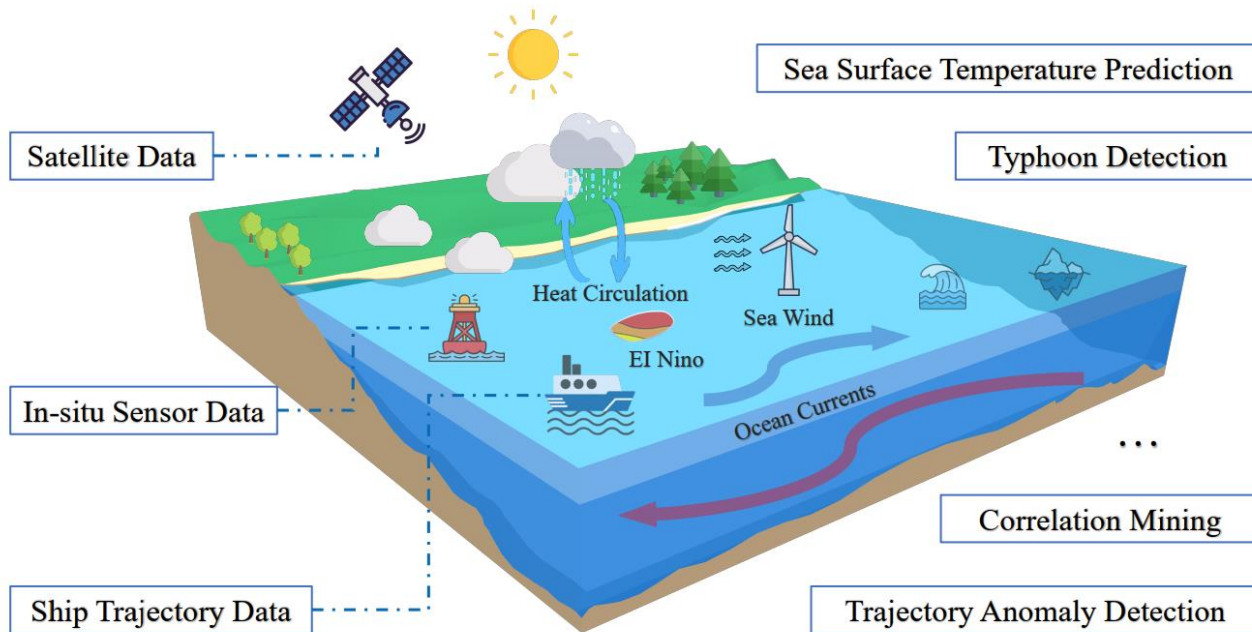
- **第15周提交**

时空数据挖掘方法

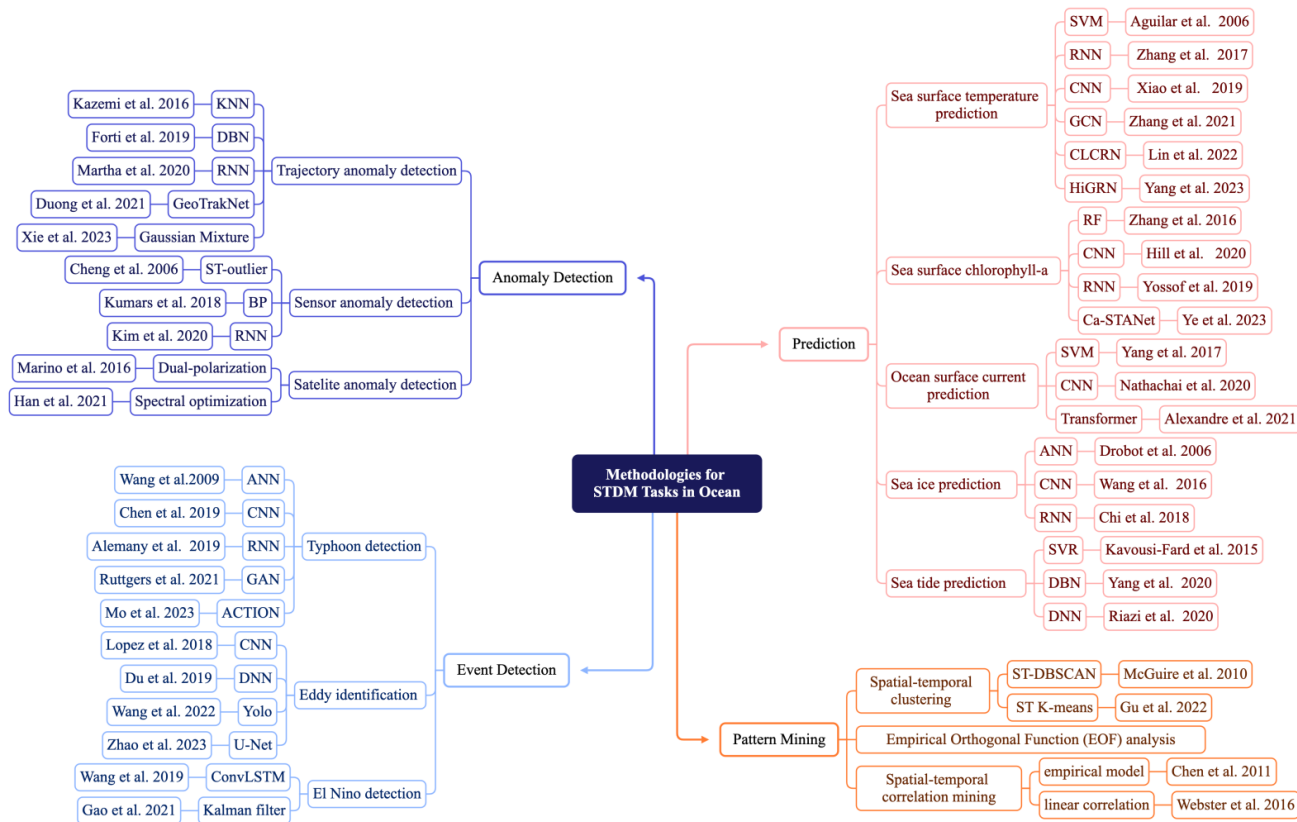
Data

STDM Methods

Tasks

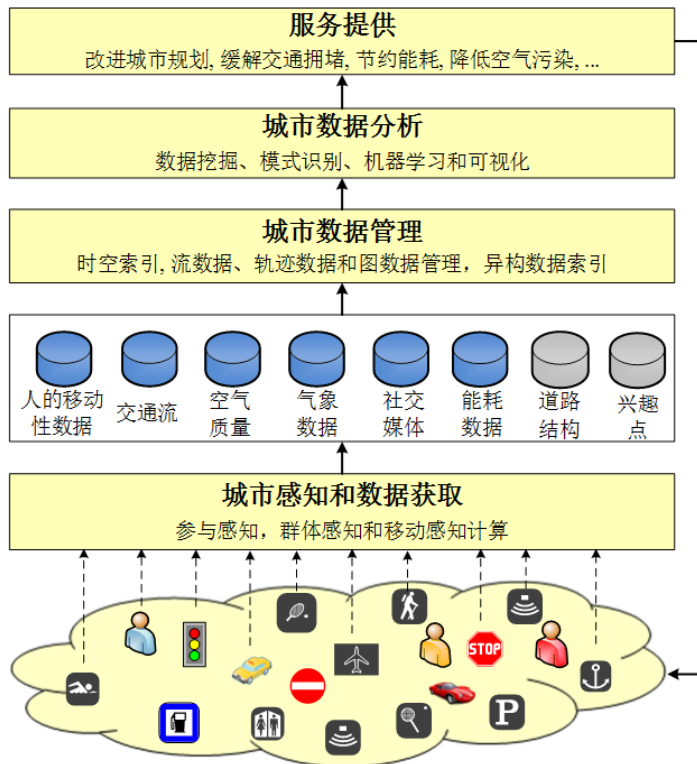


海洋计算：主要研究问题



Category	Name	Period	Spatial Resolution	Coverage	Temporal Resolution	Citation	Type	Source
Satellite Data	MODIS	2000 to present	0.041x0.041	Global	8 days	[81, 84, 107, 108]	Sea surface temperature, ocean color, sea surface salinity	https://modis.gsfc.nasa.gov/
		2002 to present	1 km x 1 km	Global	daily	[191]		
		2002 to present	0.083°x 0.083°	Global	monthly	[97, 98]		
	AVHRR	1979 to present	1.1 km x 1.1 km	Global	daily	[74, 107]	Sea surface temperature, ocean color	https://www.eumetsat.int/avhrr
	Sentinel-3	2016 to present	1.2 km x 1.2 km	Global	5 days	[20, 21, 49, 83, 121, 125, 217]	Sea surface temperature, ocean color	https://sentinels.copernicus.eu/web/sentinel/
	GOCI	2010 to 2021	0.5 km x 0.5 km	Korean sea	hourly	[126, 158, 187, 226]	Sea surface chlorophyll-II, ocean color	https://oceancolor.gsfc.nasa.gov/data/goci/
	CZCS	1978-1986	0.825 km x 0.825 km	Global	8 days	[141, 148, 162, 231]	Sea surface chlorophyll-II	https://oceancolor.gsfc.nasa.gov/data/CZCS/
	OCM-2	2009 to present	1-4 km x 1-4 km	Global	2 days	[162]	Ocean color	https://ioccc.org/sensor/ocm-2/
In-situ data	SeaWiFS	1997-2010	1-4 km x 1-4 km	Global	daily	[31, 114, 141, 203]	Ocean color	https://oceancolor.gsfc.nasa.gov/SeaWiFS/
	Argo	1996-present	Trajectories of about 14060 floats	Global	1-10 days	[12, 17, 28, 212, 218]	sea surface temperature, sea surface salinity	https://argo.ucsd.edu/
	SOCCOM	2004-present	Trajectories of about 200 floats	Antarctic Ocean	10 days	[26, 92]	Ocean carbon	https://socc.com.princeton.edu/
Ships Data	GO-BGC	2021-present	Trajectories of about 500 floats	Global	10 days	[25, 79, 182, 216]	Sea O ₂ , sea Ph	https://www.go-bgc.org/
	AIS	2016 to 2018	Trajectories of about 70,000 vessels	Global	30 seconds - 1 day	[23, 45, 53, 82]	Trajectory anomalies, ship tracking	https://www.vmsdata.com/
Reanalysis Data	VMS	April, 2020	Trajectories of 750,000 vessels	Global	30 seconds - 1 day	[60, 177, 185]	Trajectory anomalies	https://marinecadastre.gov/
	OISST	1979-present	0.25°x0.25°	Global	daily	[86, 214, 234, 241]	Sea surface temperature	https://www.ncei.noaa.gov
	ERA-5	1959-present	4°x4°	Global	12-hour	[111, 128, 140, 233]	Sea surface temperature	https://www.ecmwf.int
	CMEMS Level 3 SLA	2004 to present	0.125°x0.125°	Global	daily	[11, 55]	Sea level anomalies	https://marine.copernicus.eu/
	CMEMS	1993-2020	0.25°x0.25°	Global	daily	[11]	Sea surface height anomaly	https://marine.copernicus.eu/
	HadCRUT4	1961-1990	5°x5°	Global	monthly	[36, 146, 176]	Air/Marine temperature anomalies	https://www.metoffice.gov.uk/
	COBE SST	1891 to present	1°x1°	Global	monthly	[33, 71]	Sea surface temperature	https://psl.noaa.gov/data/
	COBE-SST 2 and Sea Ice	1850 to 2019	1°x1°	Global	monthly	[189], [33]	Sea surface temperature, sea ice concentration	https://psl.noaa.gov/data/
	CMAP Precipitation	1979 to present	2.5°x2.5°	Global	monthly	[40, 102, 119]	Pentad global gridded precipitation means	https://psl.noaa.gov/data/
	WOD	1772 to 2017	1°x1°	Global	daily	[22]	Sea temperature, salinity, oxygen	https://www.ncei.noaa.gov/products/world-ocean-database

城市计算(Urban Computing)



城市计算



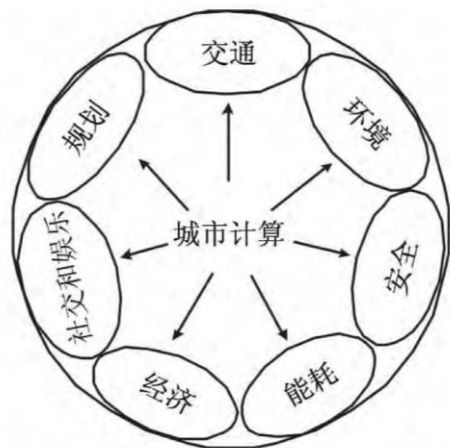
同济大学
TONGJI UNIVERSITY

- [1] 百度城市大脑白皮书
- [2] 阿里巴巴区块链赋能新型智慧城市白皮书
- [3] 京东云智能城市白皮书
- [4] 中国智能城市发展战略与策略研究
- [5] 城市交通数字化转型白皮书
- [6] 赛迪白皮书：2020城市新基建布局与发展白皮书

百度AI 新基建版图







智慧基建



智慧环保



智慧交通



智慧能耗



智慧安防

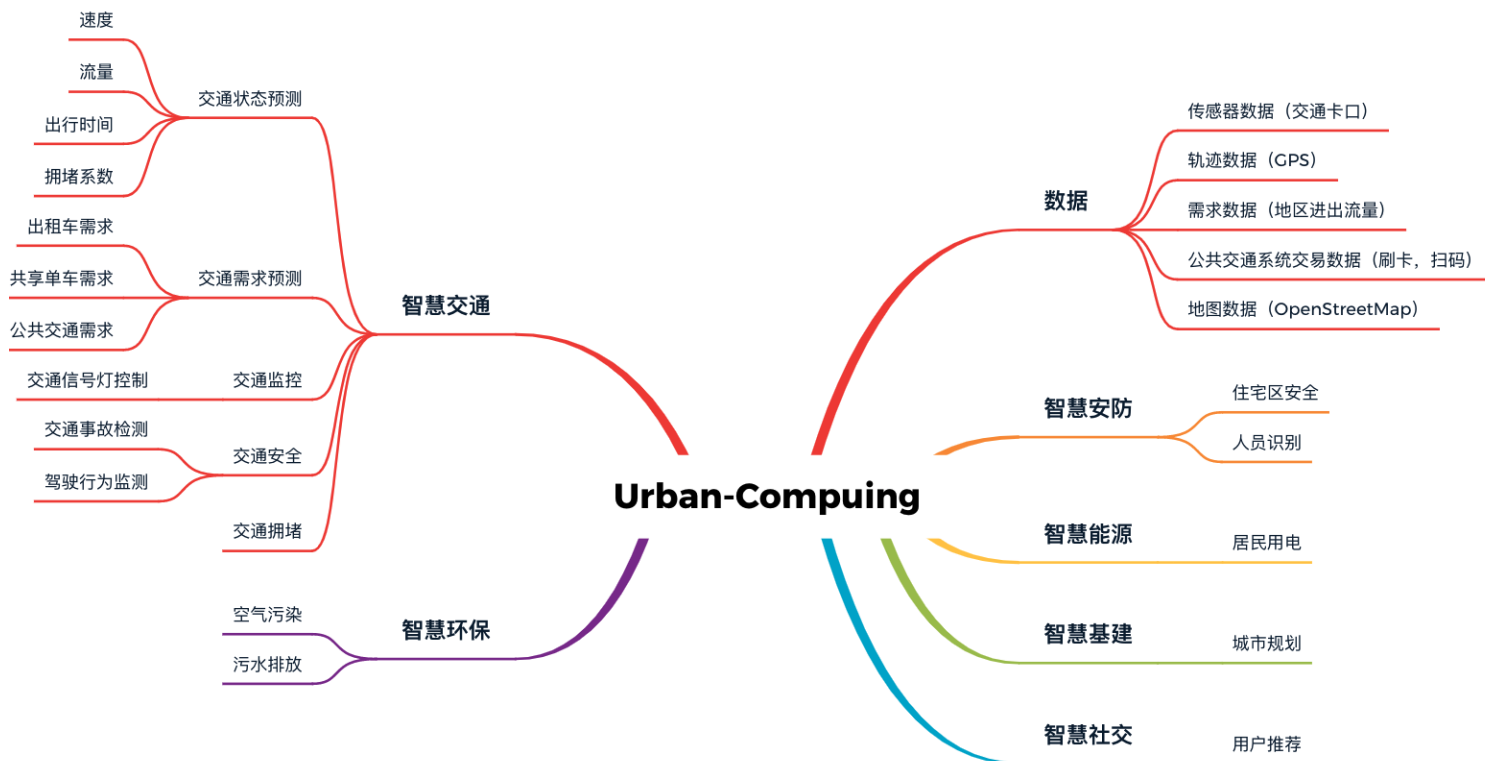


智慧社交

城市计算：主要研究问题



同济大学
TONGJI UNIVERSITY



- **国内城市数据**

- 上海市公共数据开放平台: <https://data.sh.gov.cn/>
- 北京市公共数据开放平台: <https://data.beijing.gov.cn/>
- 深圳市政府数据开放平台: <https://opendata.sz.gov.cn/>

- **国外城市数据**

- 纽约城市数据: <https://opendata.cityofnewyork.us/data/>
- 芝加哥城市数据: <https://data.cityofchicago.org/>

- **其他数据**

- OpenStreetMap地图数据: <https://download.geofabrik.de/>
- 百度地图、高德地图 (API接口获取)